The Dampened Automation Buffer: Monetary Policy and Household Heterogeneity*

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Abstract

This paper embeds endogenous automation into a Two-Agent New Keynesian (TANK) model to analyze how restricted financial participation alters monetary transmission. In representative-agent frameworks, contractionary shocks induce an aggressive "de-automation buffer" as firms substitute machines for workers, generating a medium-term employment boom. I show that this outcome is a result of the representative-agent assumption. Breaking the feedback loop between corporate profits and worker income causes firms to de-automate significantly less. Consequently, standard models overstate labor substitution during a monetary tightening, severely underestimating the persistence of labor market slack during the recovery.

Keywords: Monetary Policy, Endogenous Automation, Household Heterogeneity, Business Cycles, Employment Dynamics

JEL codes: E12, E32, E52, O33

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1 Introduction

Endogenous automation introduces a novel margin for monetary policy: the firm's choice between machines and human labor. In standard Representative-Agent New Keynesian (RANK) frameworks, a monetary contraction raises the cost of capital, inducing firms to "de-automate" and substitute machines with workers (Wolf and Fornaro, 2021). This shift acts as a macroeconomic buffer, softening the initial decline in aggregate employment. However, this mechanism implies that during the recovery phase, employment can rebound past its steady state, generating a medium-term labor market "boom" even as output remains depressed.

This prediction sits in tension with standard empirical assessments of monetary policy shocks, which document a persistent decline in labor market activity with no evidence of a medium-term overshoot (Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021). Whether this discrepancy reflects misspecification of the household structure, other modeling omissions, or the limited empirical relevance of the automation margin is an open question. In this work I address the first possibility.

This paper argues that this labor buffer is a result of the representative-agent assumption. By assuming a single household both supplies labor and owns all corporate equity, RANK models inherently conflate wage income with corporate profits. I embed endogenous automation into a Two-Agent New Keynesian (TANK) model to break this unrealistic feedback loop. I depart from the standard benchmark by dividing the economy into "capitalists," who accumulate physical capital and absorb dividend fluctuations, and "workers," who rely on wage income.

The separation of dividend income from wage income has a fundamental consequence for the transmission of monetary shocks. I show that restricted financial participation fundamentally alters monetary policy transmission through a wage-wealth channel. In RANK, the profit collapse raises the representative household's marginal utility of consumption. The household therefore shifts labor supply outward, accepting an aggressive wage decline. Since the automation threshold is governed by the ratio of the real wage to the rental rate of capital, this wage collapse induces aggressive de-automation. In

TANK, workers are insulated from the dividend collapse, their marginal rate of substitution remains anchored to labor income alone, and wages stay comparatively elevated. The analysis therefore reveals that standard representative-agent frameworks overstate labor substitution by amplifying the subsequent recovery.

2 The Model

The model is composed of households, a production sector, and a central bank. This section outlines the key features of the framework to provide the main intuition. The full set of equations and derivations is reported in Appendix A, while the parameter calibration is presented separately in Appendix B.

2.1 Households

Following Cantore and Freund (2021), households are divided into capitalists and workers. Capitalists do not supply labor; they own the production sector, receive all corporate dividends, and accumulate physical capital subject to standard utilization and investment adjustment costs. Workers, conversely, supply differentiated labor services subject to a Rotemberg-type nominal wage adjustment cost and consume out of their wage income. This feature is essential for generating pro-cyclical dividends in DSGE models, a property that becomes particularly important in frameworks with household heterogeneity.[1] Crucially, workers do not participate in financial markets and receive no dividend income. This TANK structure breaks the standard assumption where a single representative household absorbs both wage and profit fluctuations.

2.2 Production Sector

The production sector incorporates the task-based framework of Acemoglu and Restrepo (2018), following the New Keynesian implementation by Fueki et al. (2023). As is standard in the New Keynesian literature, final-good producers aggregate differentiated in-

[1]For a detailed discussion of the implications of counter-cyclical dividends in heterogeneous-agent models, see Broer et al. (2020).

termediate goods into a single final good, while intermediate-goods producers operate under monopolistic competition with Rotemberg price stickiness. The novel mechanism of the model emerges at the level of task production.

A perfectly competitive task aggregator f produces a unique aggregated task $y_t(f)$ by combining a unit measure of tasks $y_{f,t}(i)$ according to the production function:

$$y_t(f) = \zeta \left(\int_{N-1}^{N} y_{f,t}(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}, \tag{1}$$

where $i \in [N-1, N]$ with N denoting the range of tasks, $\zeta > 0$ is a scale parameter, and σ represents the elasticity of substitution across tasks.

Task producers also operate under perfect competition. Each task producer i supplies a unique task $y_{f,t}(i)$ to the task aggregator f, using capital services $k_{f,t}(i)$ and labor $l_{f,t}(i)$. Following Acemoglu and Restrepo (2018), a technological constraint on automation is present, $A \in [N-1, N]$, such that tasks $i \le A$ are technologically automated and can therefore be produced with capital, while the remaining tasks are not automated and must be produced with labor. The production function of task producer i is given by:

$$y_{f,t}(i) = \begin{cases} k_{f,t}(i) + \gamma(i) l_{f,t}(i) & (i \le A) \\ \gamma(i) l_{f,t}(i) & (i > A) \end{cases}, \tag{2}$$

where $\gamma(i)$ denotes labor productivity in task i. I assume that $\gamma(i)$ is strictly increasing in i, which implies that human labor has a strict comparative advantage in tasks with a higher index.

While the full profit maximization problem for task producers is presented in Appendix A, it leads to the following condition:

$$r_t^K = \frac{w_t}{\exp(\mu A_t^*)}, \tag{3}$$

where r^K and w are real rental cost of capital and real wage, respectively, and $\mu > 0$ is a parameter governing the curvature of $\gamma(i)$. Rearranging (3), we obtain:

$$A_t^* = \frac{1}{\mu} \ln\left(\frac{w_t}{r_t^K}\right) = \frac{\ln(w_t) - \ln(r_t^K)}{\mu} \tag{4}$$

Equation (4) identifies a unique threshold task A^*, at which task producers are indifferent between using capital or labor for producing tasks with $i \leq A^*$. Following Acemoglu and Restrepo (2018), I assume that producers employ capital services whenever they are indifferent between the use of capital and labor.

Most importantly, A^* serves as a proxy for the degree of automation in production. An increase in A^* indicates a shift toward a more automated economy, since a larger share of tasks is then produced exclusively with capital. In line with Fueki et al. (2023), A^* can therefore be interpreted as the "automation rate".[2] Equation (4) thus shows that the automation rate is determined by the arbitrage between the real rental cost of capital and the real wage. This arbitrage condition is the key equation of the model: anything that differentially affects the real wage relative to the rental rate of capital will alter the degree of automation.

2.3 Central Bank

The central bank sets the nominal interest rate according to a standard Taylor rule with interest rate inertia, subject to exogenous monetary policy shocks.

3 Results

To illustrate the implications of restricted financial participation, Figure 1 contrasts the impulse responses of the TANK and RANK models to a 25-basis-point monetary contraction.[3] Along standard macroeconomic dimensions—output, investment, and inflation—the two frameworks are broadly similar, generating standard recessionary dynamics. Even the rental rate of capital follows similar trajectories. This equivalence sharpens

[2] Note that Fueki et al. (2023) study two cases: one in which the technological constraint on automation does not bind, and another in which an upper bound exists. I abstract from this distinction and consider only the case in which the automation rate is not subject to any upper-bound limitation.

[3] Apart from the specification of household heterogeneity, the calibration is identical across the two model variants.

the focus on their critical point of divergence: the real wage and the automation rate.

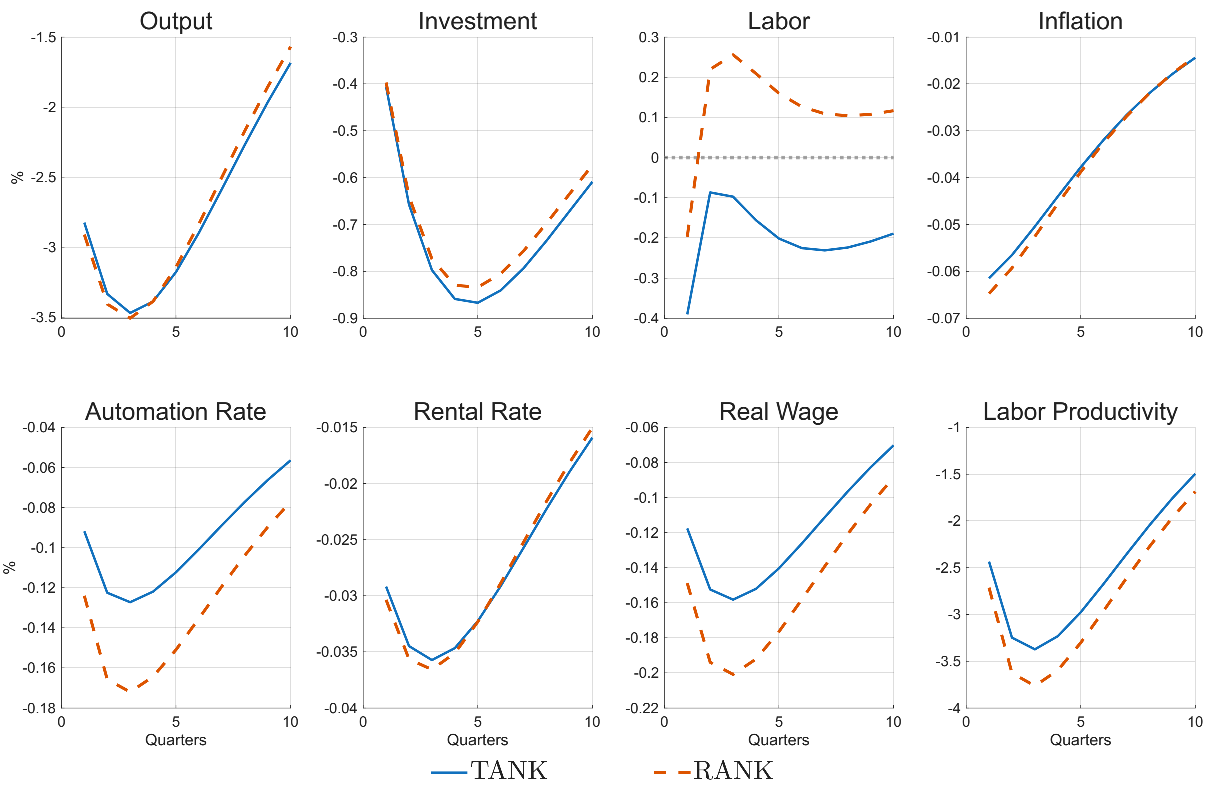

Figure 1: Impulse response to a monetary contraction for aggregate variables

Note: Increase in the nominal interest rate of 25 b.p. The blue solid line refers to an economy with heterogeneous households. The red dashed line refers to the case of a representative household.

To better understand these distinct automation dynamics, we can decompose the automation choice. As established in equation (4), the optimal automation threshold is dictated by the relative cost of factors. Taking a first-order Taylor approximation around the steady state yields:

$$dA_t^* \approx \frac{1}{\mu}\hat{w}_t - \frac{1}{\mu}\hat{r}_t^K \tag{5}$$

where dA_t^* represents the absolute deviation of the automation rate from its steady state, while $\hat{w}_t$ and $\hat{r}_t^K$ denote the percentage deviations of the real wage and the rental rate of capital, respectively. This linear approximation isolates the two opposing forces driving firms' automation choices following a contractionary monetary shock.

Figure 2 plots this decomposition for both the TANK and RANK economies. The positive contribution (pushing automation up) arises from the decline in the rental rate of capital, which makes machines relatively cheaper. Conversely, the negative contribution (driving de-automation) is generated by the decline in the real wage, which makes human labor relatively cheaper.

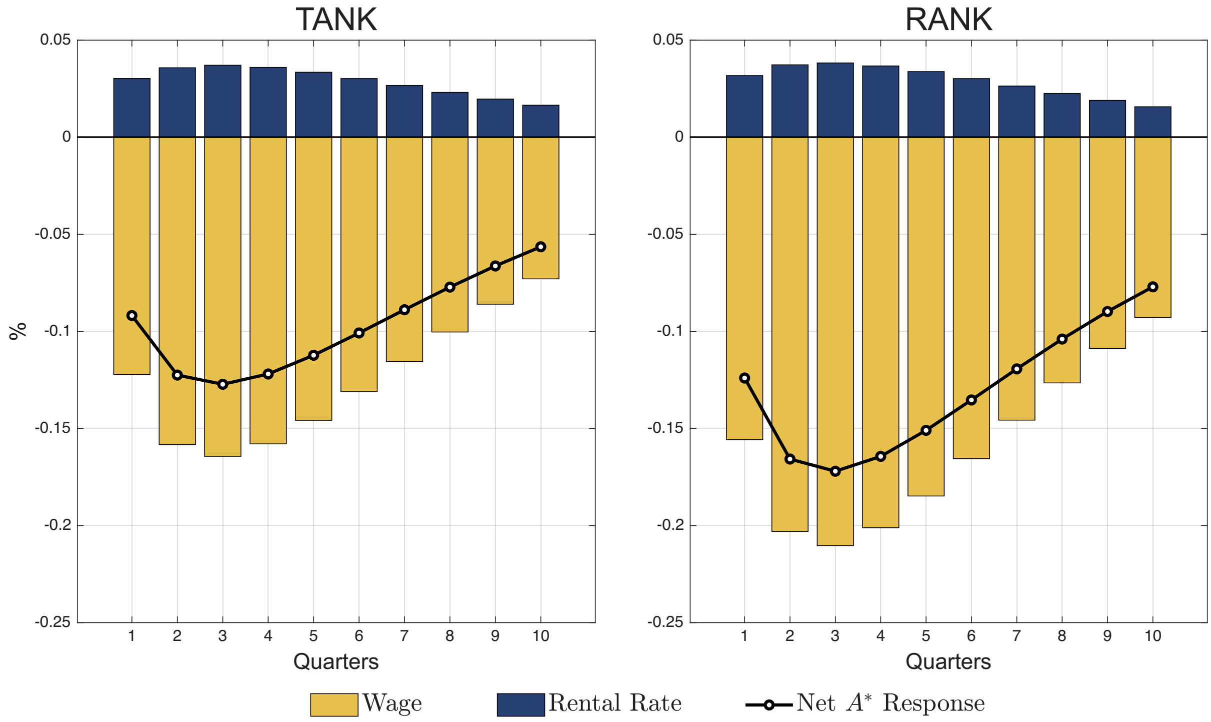

Figure 2: Decomposition of the Automation Rate Response

Note: The figure plots the first-order approximation components of the automation rate deviation following a 25 b.p. monetary contraction. Stacked bars represent the isolated contributions of the real wage ($\hat{w}_t$) and the rental rate of capital ($\hat{r}_t^K$). The solid black line with markers represents the net simulated response of the automation rate.

Because rental rate dynamics are similar across both models, the discrepancy in the net automation response is entirely driven by the wage channel. In the RANK economy, the monetary tightening causes corporate profits to collapse. The representative household absorbs this dividend loss entirely, triggering a strong negative wealth effect that shifts labor supply outward. The household accepts a sharp real wage decline, making human labor relatively cheaper. Firms exploit this by aggressively substituting machines for workers. This "de-automation buffer" cushions the initial employment drop but generates an medium-term employment boom during the recovery phase, even while output remains depressed.

The TANK framework breaks this mechanism. Because workers do not receive dividends, they are insulated from the profit collapse. Their effective labor supply does not shift outward, and wages remain relatively elevated. Facing a more expensive labor input, firms de-automate significantly less. Stripped of the de-automation buffer, the economy experiences an initial employment contraction that is strictly deeper than in the RANK baseline, followed by a highly persistent decline with no medium-term overshoot.

Despite the concentration of the entire dividend contraction on a small group of capitalists, aggregate investment dynamics in the TANK model remain broadly similar to those in the RANK benchmark, albeit with a more persistent decline. When dividend income collapses, capitalists choose to issue debt to workers, who retain a higher capacity to save because they are insulated from the profit shock, rather than reduce capital accumulation, which is strongly penalized by standard investment adjustment costs. The concentrated wealth shock is therefore absorbed through intra-household borrowing and greater volatility in capitalist consumption, leaving the aggregate investment trajectory substantially unaffected. As a result, the two models deliver fairly similar results in terms of output, investment, and inflation, yet they generate sharply different employment paths. Household heterogeneity, operating through the automation margin, thus emerges as a powerful source of hidden misspecification in standard frameworks: one that is invisible in the aggregates typically used to assess monetary transmission but critical for understanding the labor market dynamics after a monetary tightening.

4 Concluding Remarks

This paper demonstrates that when automation is an endogenous choice, the distribution of financial assets is crucial for monetary transmission on employment. While aggregate output and inflation dynamics remain substantially invariant to household heterogeneity, the underlying adjustments in factor markets tell a different story. Household heterogeneity weakens the de-automation buffer observed in representative-agent models by breaking the link between corporate profits and worker wealth. As a result, limited financial participation substantially modifies the employment dynamics of monetary policy by intensifying the initial impact and prolonging the overall contraction in labor.

For macroeconomic modeling, the implication is that standard frameworks overstate firms' willingness to substitute toward human labor during monetary tightenings, generating an overly optimistic employment rebound that understates the broader costs of contractionary shocks. To the extent that financial participation is restricted in practice, these frameworks may mischaracterize the persistence of labor market slack following a monetary tightening. Whether this theoretical mechanism is quantitatively significant remains a question for future empirical work.

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Appendix

A Model Details

This appendix section provides the complete model introduced in Section 2. To build it, I adopt the simplest possible framework in order to maximize transparency and focus exclusively on the qualitative implications of household heterogeneity for automation. Despite its simplicity, the model remains rich in features, given the nature of the analysis.

Within the model, expressions such as $X_{x,t}$ indicate per-capita variables, whereas X_t refers to aggregate values. Moreover, $\bar{X}$ represents the steady-state level of the corresponding variable.

The economy consists of three main blocks: households, firms (that is, the production sector), and a central bank. Time is discrete and infinite. The behavior of each agent is described in detail below.

A.1 Households

Household heterogeneity is incorporated in the simplest possible way. Following Cantore and Freund (2021), I rely on a TANK structure that separates households into two groups, capitalists (C) and workers (W).[4]

A.1.1 Capitalists

Capitalists do not supply labor and derive utility exclusively from consumption. The utility function for a representative capitalist j is:

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{C_{c,t}(j)^{1-\xi}}{1-\xi} , \tag{A1}$$

where $C_{c,t}(j)$ is the capitalist consumption, β is the discount factor, and ξ is the coefficient of relative risk aversion.

Capitalist j can save (or borrow) in liquid bonds, rent capital to firms, and earn firms'

[4] A similar structure is also employed in Klein and Krause (2020).

dividends, given her ownership of the production sector. In addition, she undertakes capital stock accumulation and capital refurbishing. Therefore, her budget constraint is:

$$
\begin{aligned}
C_{c,t}(j) + B_{c,t}(j) + I_{c,t}(j) + \Psi\left\{u_{c,t}(j)\right\} \tilde{K}_{c,t-1}(j) = \\
r_t^K u_{c,t}(j) \tilde{K}_{c,t-1}(j) + \frac{R_{t-1}}{\pi_t} B_{c,t-1}(j) + D_{c,t}(j) + \frac{Tr}{\lambda} \,,
\end{aligned} \tag{A2}
$$

where $B_{c,t}(j)$ denotes the level of bonds in capitalist j's portfolio, $I_{c,t}(j)$ represents investment expenditures in capital, $\tilde{K}_{c,t}(j)$ is the capital stock holding, $u_{c,t}(j)$ is the capital utilization rate, $D_{c,t}(j)$ are dividends stemming from firms' operations, Tr represents the transfers between household types introduced to ensure zero consumption inequality in the steady state, and λ is the share of capitalists among households. r^K and R denote the real interest rate on capital and the nominal interest rate on bonds, respectively, while π is the inflation rate, defined later in Section A.2.2.

The term $\Psi\left\{u_{c,t}(j)\right\}$ is a capital utilization cost function which, in line with Christiano et al. (2010), is defined as:

$$
\Psi(u_{c,t}(j)) = 0.5\, \sigma_a \sigma_b u_{c,t}(j)^2 + \sigma_b (1 - \sigma_a) u_{c,t}(j) + \sigma_b \left(\frac{\sigma_a}{2} - 1 \right) \,, \tag{A3}
$$

where σ_a is the parameter governing the curvature of the cost function, and σ_b is calibrated such that $\Psi(u) = \Psi'(u) = 0$ at the steady state.

Since capitalist j undertakes capital stock accumulation and capital refurbishing, she is also subject to the following law of motion for capital:

$$
\tilde{K}_{c,t}(j) = (1 - \delta) \tilde{K}_{c,t-1}(j) + \left\{ 1 - \frac{\phi}{2} \left(\frac{I_{c,t}(j)}{I_{c,t-1}(j)} - 1 \right)^2 \right\} I_{c,t}(j) \,, \tag{A4}
$$

where δ is the depreciation rate of capital, and ϕ denotes the investment adjustment costs.

Capitalist j rents the capital service $K_{c,t}(j)$ to firms at the rental rate r^K, and she can adjust the amount of capital service by choosing the utilization rate $u_{c,t}(j)$. The capital service is therefore defined as:

$$K_{c,t}(j) = u_{c,t}(j)\tilde{K}_{c,t-1}(j) \ . \tag{A5}$$

Capitalist j chooses $C_{c,t}(j)$, $B_{c,t}(j)$, $I_{c,t}(j)$, $u_{c,t}(j)$, and $\tilde{K}_{c,t}(j)$ to maximize the expected lifetime utility given in equation (A1), subject to equations (A2) and (A4). Since all capitalists behave identically, I can ignore the superscript j and derive the following First Order Conditions (FOC):

- w.r.t. $C_{c,t} \rightarrow$ Lagrangian multiplier for capitalist consumption:

$$\Lambda_{c,t} = C_{c,t}^{-\xi} \ . \tag{A6}$$

- w.r.t. $B_{c,t} \rightarrow$ Capitalist Euler equation:

$$\Lambda_{c,t} = \beta E_t \left[\Lambda_{c,t+1} \frac{R_t}{\pi_{t+1}} \right] \ . \tag{A7}$$

- w.r.t. $I_{c,t} \rightarrow$ Optimality condition for investment

$$\begin{aligned} q_t & \left\{ 1 - \phi \left(\frac{I_{c,t}}{I_{c,t-1}} - 1 \right) \frac{I_{c,t}}{I_{c,t-1}} - \frac{\phi}{2} \left(\frac{I_{c,t}}{I_{c,t-1}} - 1 \right)^2 \right\} \\ &= 1 - E_t \left[\Omega^c_{t,t+1} q_{t+1} \phi \left(\frac{I_{c,t+1}}{I_{c,t}} - 1 \right) \left(\frac{I_{c,t+1}}{I_{c,t}} \right)^2 \right] \ , \end{aligned} \tag{A8}$$

where q is Tobin's q, derived from the Lagrange multiplier associated with the constraint (A4), and $\Omega^c_{t,t+1} = \beta\left(\Lambda_{c,t+1}/\Lambda_{c,t}\right)$ is the stochastic discount factor for capitalists.

- w.r.t. $\tilde{K}_{c,t} \rightarrow$ Optimality condition for capital stock:

$$q_t = E_t \left[\Omega^c_{t,t+1} \left\{ u_{c,t+1} r^K_{t+1} - \Psi(u_{c,t+1}) + (1-\delta) q_{t+1} \right\} \right] \ . \tag{A9}$$

- w.r.t. $u_t \rightarrow$ Optimality condition for utilization rate:

$$r_t^K = \sigma_a \sigma_b (u_{c,t} - 1) + \sigma_b \; . \tag{A10}$$

A.1.2 Workers

Workers derive utility from consumption and experience disutility from labor. The utility function for a representative worker j is:

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{C_{w,t}(j)^{1-\xi}}{1-\xi} - \frac{L_{w,t}(j)^{1+\eta}}{1+\eta} \; , \tag{A11}$$

where $C_{w,t}(j)$ is worker consumption, β is the discount factor, and ξ is the coefficient of relative risk aversion (the last two parameters are identical across workers and capitalists). $L_{w,t}(j)$ denotes the labor supplied by worker j, and η is the inverse Frisch elasticity of labor supply.

Worker j earns labor income from working and can also borrow or save in bonds. I assume that workers supply differentiated labor services and face a Rotemberg-type nominal wage adjustment cost, which introduces wage stickiness. Therefore, her budget constraint is:

$$\begin{aligned} C_{w,t}(j) + B_{w,t}(j) + \frac{\psi_w}{2}\left(B_{w,t}(j) - \bar{B}_w(j)\right)^2 = {} & w(j)_t L_{w,t}(j) \left[1 - \frac{\phi_w}{2}\left(\pi_t^w - 1\right)^2\right] \\ & + \frac{R_{t-1}}{\pi_t} B_{w,t-1}(j) + f_t - \frac{Tr}{1-\lambda} \; , \end{aligned} \tag{A12}$$

where $B_{w,t}(j)$ denotes the level of bonds in worker j' portfolio, $w(j)$ is her real wage, and Tr represents the transfers between household types introduced to ensure zero consumption inequality in the steady state. The term $\frac{\psi_w}{2}\left(B_{w,t}(j) - \bar{B}_w(j)\right)^2$ is a convex portfolio adjustment cost function which, following Schmitt-Grohé and Uribe (2003), is introduced to ensure stationarity. The parameter ψ_w therefore captures the strength of this adjustment cost. Following Cantore and Freund (2021), and to rule out any wealth effects, these costs are rebated to workers as a lump-sum payment, f_t, which workers do not internalize when making their savings decisions.

To introduce sticky wages, I assume that worker j supplies differentiated labor services, $L_{w,t}(j)$, according to the following labor demand equation:

$$L_{w,t}(j) = \left(\frac{W_t(j)}{W_t}\right)^{-\epsilon_w} L_t \ , \tag{A13}$$

where $W_t(j)$ is the nominal wage of worker j, W_t is the nominal aggregate wage, L_t is aggregate labor, and ϵ_w is the elasticity of substitution across differentiated labor services. The term $(\phi_w/2)\left(\pi_t^w - 1\right)^2$ in the worker budget constraint represents the nominal wage adjustment cost à la Rotemberg (1982), where ϕ_w is the parameter governing the strength of this adjustment cost. The variable π^w represents the nominal wage inflation rate, defined as: $\pi_t^w = W_t(j)/W_{t-1}(j) = \left[w_t(j)/w_{t-1}(j)\right]\pi_t$

Worker j chooses $C_{w,t}(j)$, $B_{w,t}(j)$, and $W_t(j)$ to maximize the expected lifetime utility specified in equation (A11), subject to equations (A12) and (A13). Since all workers behave identically, I can omit the superscript j and derive the following First Order Conditions (FOC):

- w.r.t. $C_{w,t} \rightarrow$ Lagrangian multiplier for worker consumption:

$$\Lambda_{w,t} = C_{w,t}^{-\xi} \ . \tag{A14}$$

- w.r.t. $B_{w,t} \rightarrow$ Worker Euler equation:

$$\Lambda_{w,t}\left[1 + \psi_w\left(B_{w,t} - \bar{B}_w\right)\right] = \beta E_t\left[\Lambda_{w,t+1}\frac{R_t}{\pi_{t+1}}\right] \ . \tag{A15}$$

- w.r.t. $W_t(j) \rightarrow$ Wage NKPC:

$$\begin{aligned} \epsilon_w \frac{1}{w_t}\frac{L_{w,t}^{\eta}}{\Lambda_{w,t}} - \left(\frac{\phi_w}{2}\left\{(3-\epsilon_w)(\pi_t^w)^2 - 2(2-\epsilon_w)\pi_t^w + (1-\epsilon_w)\right\} - (1-\epsilon_w)\right) \\ + E_t\left[\Omega_{t,t+1}^w \phi_w\left\{\pi_{t+1}^w - 1\right\}\pi_{t+1}^w\left(\frac{L_{w,t+1}}{L_{w,t}}\right)\right] = 0 \ , \end{aligned} \tag{A16}$$

where $\Omega_{t,t+1}^w = \beta\left(\Lambda_{w,t+1}/\Lambda_{w,t}\right)$ is the stochastic discount factor for workers.

A.2 Production Sector

The production sector incorporates the production-task approach of Acemoglu and Restrepo (2018) into a DSGE framework, following the specification of Fueki et al. (2023). This part of the economy is populated by final-good producers, intermediate-good producers, task aggregators, and task producers.

A.2.1 Final-Good Producers

Final-good producers purchase differentiated intermediate goods, $Y_t(f)$, $f \in [0,1]$, to produce a single final good, Y_t, according to the following production function:

$$Y_t = \left(\int_0^1 Y_t(f)^{\frac{\epsilon_p - 1}{\epsilon_p}} df \right)^{\frac{\epsilon_p}{\epsilon_p - 1}}, \tag{A17}$$

where ϵ_p denotes the elasticity of substitution across intermediate goods. The final-good producer solves the following profit maximization problem:

$$\max_{Y_t(f)} \quad P_t Y_t - \int_0^1 P_t(f) Y_t(f) df \ , \tag{A18}$$

subject to equation (A17), and where P_t and $P_t(f)$ denote the prices of the final good and of intermediate good f, respectively. Profit maximization leads to the following demand function for intermediate good f:

$$Y_t(f) = \left(\frac{P_t(f)}{P_t} \right)^{-\epsilon_p} Y_t. \tag{A19}$$

A.2.2 Intermediate-Good Producers

Intermediate-good producer f supplies a unique intermediate good, $Y_t(f)$ under monopolistic competition, by employing an aggregated task, $y_t(f)$, at price $p_t(f)$. Its production function is:

$$Y_t(f) = y_t(f) \ . \tag{A20}$$

I assume that she faces a Rotemberg-type price adjustment cost, which introduces price stickiness into the economy. Therefore, the profit maximization problem of producer f is the following:

$$\max_{P_t(f)} E_t \sum_{k=0}^{\infty} \Omega^c_{t,t+k} \left[P_{t+k}(f) Y_{t+k}(f) - p_{t+k}(f) y_{t+k}(f) - \frac{\phi_p}{2} \left(\pi_t(f) - 1\right)^2 P_{t+k} Y_{t+k} \right] , \tag{A21}$$

where $\Omega^c_{t,t+k}$ denotes the stochastic discount factor of capitalists, who own the entire production sector, and the term $(\phi_p/2)\left(\pi_t(f) - 1\right)^2$ captures the adjustment costs introduced by Rotemberg (1982). The parameter ϕ_p measures the intensity of these costs, and price inflation for intermediate good f is defined as $\pi_t(f) = P_t(f)/P_{t-1}(f)$.

All intermediate-goods firms face the same optimization problem, which leads them to choose identical prices and quantities. Consequently, the first-order condition of the profit maximization problem yields the nonlinear price NKPC:

$$(1 - \epsilon_p) + \epsilon_p MC_t - \phi_p(\pi_t - 1)\pi_t + \phi_p E_t \left[\Omega^c_{t,t+1} (\pi_{t+1} - 1) \pi_{t+1} \frac{Y_{t+1}}{Y_t} \right] = 0 , \tag{A22}$$

where MC is the real marginal cost, defined as $MC_t = p_t/P_t$.

A.2.3 Task Aggregators

A perfectly competitive task aggregator f produces a unique aggregated task $y_t(f)$ by combining a unit measure of tasks $y_{f,t}(i)$ according to equation (1)

The profit maximization problem of the task aggregator f is the following:

$$\max_{y_{f,t}(i)} \quad p_t(f) y_t(f) - \int_{N-1}^{N} p_{f,t}(i) y_{f,t}(i) di , \tag{A23}$$

subject to the equation (1). Therefore, the demand function for task i is:

$$y_{f,t}(i) = \zeta^{\sigma - 1} \left(\frac{p_{f,t}(i)}{p_t(f)} \right)^{-\sigma} y_t(f) , \tag{A24}$$

and price of aggregated task, $p_t(f)$, is given by:

$$p_t(f) = \frac{1}{\zeta}\left(\int_{N-1}^{N} p_{f,t}(i)^{1-\sigma} di\right)^{\frac{1}{1-\sigma}} . \tag{A25}$$

A.2.4 Task Producers

Task producers also operate under perfect competition. Task producer i supplies a unique task $y_{f,t}(i)$ to the task aggregator f, using capital services $k_{f,t}(i)$ and labor $l_{f,t}(i)$. Following Acemoglu and Restrepo (2018), a technological constraint on automation is present, $A \in [N-1, N]$, such that tasks $i \leq A$ are technologically automated and can therefore be produced with capital, while the remaining tasks are not automated and must be produced with labor. The production function of task producer i is given by equation (2)

Task producer i, therefore, solves two different problems, depending on whether $i \leq A$:

Case	Optimization Problem	Constraint
$i \leq A$	$\max\limits_{\{k_{f,t}(i), l_{f,t}(i)\}} p_{f,t}(i) y_{f,t}(i) - \left(R_t^K k_{f,t}(i) + W_t l_{f,t}(i)\right)$	$y_{f,t}(i) = k_{f,t}(i) + \gamma(i) l_{f,t}(i)$
$i > A$	$\max\limits_{l_{f,t}(i)} p_{f,t}(i) y_{f,t}(i) - W_t l_{f,t}(i)$	$y_{f,t}(i) = \gamma(i) l_{f,t}(i)$

where R^K is the nominal rental cost of capital. Since all the tasks are produced in a perfectly competitive manner, their prices are equal to the minimum unit cost of production:

$$p_{f,t}(i) = \begin{cases} \min\left\{R_t^K, \frac{W_t}{\gamma(i)}\right\} & (i \leq A) \\ \frac{W_t}{\gamma(i)} & (i > A) \end{cases} . \tag{A26}$$

Following Acemoglu and Restrepo (2018), the productivity of labor in task i is specified as:

$$\gamma(i) = \exp(\mu i) , \tag{A27}$$

with $\mu > 0$. This formulation leads directly to the fundamental equation (4) for the automation rate presented in the main text.

A.2.5 Aggregation

As stated in Section A.2.2, all intermediate-good firms make identical decisions, which allows me to drop the subscript f.

By combining equations (A20), (A24), (2), and (A26), I obtain the task-level capital demand and labor demand:

$$k_t(i) = \begin{cases} \zeta^{\sigma-1} Y_t R_t^{K^{-\sigma}} p_t^{\sigma} & (i \leq A_t^*) \\ 0 & (i > A_t^*) \end{cases}, \qquad l_t(i) = \begin{cases} 0 & (i \leq A_t^*) \\ \zeta^{\sigma-1} Y_t \frac{1}{\gamma(i)} \left(\frac{W_t}{\gamma(i)}\right)^{-\sigma} p_t^{\sigma} & (i > A_t^*) \end{cases}.$$

Combining these equations with $MC_t = p_t/P_t$ and (A27), and aggregating across tasks, I obtain the capital and labor market clearing conditions:

$$r_t^K = MC_t \left[\zeta^{\frac{\sigma-1}{\sigma}} (A_t^* - N + 1)^{\frac{1}{\sigma}} \left(\frac{Y_t}{K_t}\right)^{\frac{1}{\sigma}} \right], \tag{A28}$$

$$w_t = MC_t \left[\zeta^{\frac{\sigma-1}{\sigma}} \left\{ \frac{e^{(\sigma-1)\mu N} - e^{(\sigma-1)\mu A_t^*}}{(\sigma-1)\mu} \right\}^{\frac{1}{\sigma}} \left(\frac{Y_t}{L_t}\right)^{\frac{1}{\sigma}} \right]. \tag{A29}$$

Combining equations $MC_t = p_t/P_t$, (A25), (A26), and (A27), I can express the real marginal cost as:

$$MC_t = \frac{1}{\zeta} \left[(A_t^* - N + 1) r_t^{K^{1-\sigma}} + \left\{ \frac{e^{(\sigma-1)\mu N} - e^{(\sigma-1)\mu A_t^*}}{(\sigma-1)\mu} \right\} w_t^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \tag{A30}$$

Combining the preceding three equations, aggregate output is given by:

$$Y_t = \zeta \left[(A_t^* - N + 1)^{\frac{1}{\sigma}} K_t^{\frac{\sigma-1}{\sigma}} + \left\{ \frac{e^{(\sigma-1)\mu N} - e^{(\sigma-1)\mu A_t^*}}{(\sigma-1)\mu} \right\}^{\frac{1}{\sigma}} L_t^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \tag{A31}$$

Finally, as is standard in New Keynesian models with Rotemberg-type price rigidity, the dividends generated by the production sector are defined as follows:

$$D_t = Y_t \left[1 - MC_t - \frac{\phi_p}{2} (\pi_t - 1)^2 \right] . \tag{A32}$$

A.3 Central bank

The central bank acts as the monetary authority and sets the nominal interest rate according to a Taylor rule with interest rate inertia and inflation targeting, defined as:

$$\frac{R_t}{\bar{R}} = \left(\frac{R_{t-1}}{\bar{R}} \right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}} \right)^{(1-\rho_R)\rho_\pi} m_t , \tag{A33}$$

where ρ_R measures the degree of interest rate inertia and ρ_π captures the sensitivity to inflation fluctuations. The exogenous shock to the nominal interest rate, m, follows an AR(1) process of the form:

$$\log(m_t) = \rho_m \log(m_{t-1}) + \epsilon_t^m , \tag{A34}$$

where ρ_m denotes the persistence of the shock, and ϵ_t^m is an i.i.d. monetary policy innovation with standard deviation σ_m.

A.3.1 Market Clearing and Aggregation

The goods market clears according to the following condition:

$$C_t + I_t + \Psi(u_t)\tilde{K}_{t-1} = \left\{ 1 - \frac{\phi_p}{2}(\pi_t - 1)^2 \right\} Y_t - \frac{\phi_w}{2}(\pi_t^w - 1)^2 w_t L_t . \tag{A35}$$

Given the absence of a bond-issuing institution, such as a government, I impose a zero-supply condition for aggregate bonds:

$$B_t = 0 . \tag{A36}$$

The aggregation relationships between aggregate and per-capita variables are reported below:

- Aggregate consumption:

$$C_t = \lambda C_{c,t} + (1 - \lambda) C_{w,t} \ . \tag{A37}$$

- Aggregate Bonds:

$$B_t = \lambda B_{c,t} + (1 - \lambda) B_{w,t} \ . \tag{A38}$$

- Aggregate Labor:

$$L_t = (1 - \lambda) L_{w,t} \ . \tag{A39}$$

- Aggregate Capital Stock:

$$\tilde{K}_t = \lambda \tilde{K}_{c,t} \ . \tag{A40}$$

- Aggregate Capital Services:

$$K_t = \lambda K_{c,t} \ . \tag{A41}$$

- Aggregate Utilization Rate:

$$u_t = u_{c,t} \tag{A42}$$

- Aggregate Investment:

$$I_t = \lambda I_{c,t} \ . \tag{A43}$$

- Aggregate Profits:

$$D_t = \lambda D_{c,t} \ . \tag{A44}$$

B Model Calibration

Table B.1 reports the parameter values used to generate the results shown in Figure 1. Most of the calibration follows standard choices in the literature, while several specific parameter values are drawn from influential related studies.

Table B.1: Baseline model calibration

Description	Param.	Value	Source/target
Households and Preferences			
Discount rate	β	0.99	Standard value
Risk aversion	ξ	2	Standard value
Inverse Frisch elasticity of labor supply	η	1	Standard value
Share of capitalists	λ	0.2	Klein and Krause (2020)
Bond portfolio adjustment cost	ψ_w	0.07	Cantore and Freund (2021)
Capital and Investment			
Depreciation rate	δ	0.025	Standard value
Investment adjustment cost	ϕ	6	Smets and Wouters (2007)
Capital utilization cost curvature	σ_a	0.01	Christiano et al. (2005)
Capital utilization parameter	σ_b	0.0351	$\Psi(u) = \Psi'(u) = 0$
Firm and Production			
Range of tasks	N	1	Normalization
Automation efficiency	μ	1	Normalization
Elasticity of substitution between tasks	σ	0.4	Chirinko and Mallick (2017)
Scale parameter	ζ	0.04	Internal calibration
Nominal Rigidities			
Elasticity of substitution: Goods	ϵ_p	6	Price markup = 20%
Elasticity of substitution: Labor	ϵ_w	6	Wage markup = 20%
Price adjustment cost (Rotemberg)	ϕ_p	29.41	Price duration = 3 quarters
Wage adjustment cost (Rotemberg)	ϕ_w	58.25	Wage duration = 4 quarters
Monetary Policy			
Interest rate smoothing	ρ_R	0.8	Clarida et al. (2000)
Taylor rule inflation coefficient	ρ_π	1.5	Standard value
Monetary shock std. dev.	σ_m	0.0025	1% annualized shock
Persistence of monetary shock	ρ_m	0	One-time innovation